

Abaad Murtaza

Data Scientist — AI & Machine Learning Engineer

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Data Scientist with 2+ years of experience in AI-driven solutions, machine learning, and backend development across research and industry. Proficient in building AI applications, including computer vision and LLM-based systems, using Python, SQL, and related technologies, with proven success in developing and deploying scalable AI solutions.

Education

Saarland University, Germany

Apr. 2024 – Present

M.Sc. Data Science and Artificial Intelligence

NED University of Engineering & Technology, Pakistan

Oct. 2018 – Oct. 2022

B.Eng. Computer Systems

Work Experience

German Research Center for Artificial Intelligence (DFKI), Germany [🔗](#)

Dec. 2024 – Present

[AI Research Assistant](#)

- Developed **text-to-BPMN** prototypes for **KicoPro** using **Llama**, **Qwen**, **GPT-OSS**, and **DeepSeek**, introducing an intermediate language (**BPIL**) that enabled around **2.5x faster** BPMN generation and used about **75% fewer tokens** than direct XML generation.
- Engineered backend and deployment pipelines for LLM-based BPMN generation using **FastAPI**, **Docker**, **Ollama**, **Hugging Face**, **Transformers**, and **vLLM**.
- Applied **prompt engineering** techniques and produced **synthetic BPMN-text datasets** for fine-tuning and evaluating LLM-based automation systems.
- Implemented a **FastAPI-based image-to-BPMN service** using **Faster R-CNN** for element detection and **DeepSeek OCR** for text extraction, converting uploaded process diagrams into BPMN models.

Bank Alfalah Limited, Pakistan [🔗](#)

Sep. 2023 – Mar. 2024

[Data Scientist](#)

- Automated OCR pipelines for verification of account statements and salary slips using **EasyOCR** and **PaddleOCR**, reducing verification time from **30 to 12 minutes** for **40+ documents daily** per officer and making the process **2.5x more efficient**.
- Built a video-based sentiment monitoring system using **DeepFace** and **Power BI** for a luxury experience branch, tracking customer counts and reactions from **4 camera feeds**.
- Collaborated with business teams on dashboards, reports, exploratory analysis, and handling **almost 2.5 TB of data** in **MS SQL Server** to support operational and risk-related decisions.

Centegy Technologies, Pakistan [🔗](#)

Sep. 2022 – Aug. 2023

[Associate Data Scientist](#)

- Implemented production-ready **Flask APIs** to integrate machine learning models into enterprise applications.
- Fine-tuned a one-shot object detector **OS2D** to automate shelf planogram compliance checks across **90+ products**, achieving **72% accuracy**.
- Optimized a truck loading solution using **3D bin packing** to improve goods arrangement, reduce planning time, and enhance space utilisation.

Skills

Languages Python, JavaScript, SQL, NoSQL

AI Tools PyTorch, HuggingFace, OpenCV, LangChain, LangGraph, Transformers, Ollama, VLLM, Numpy, Matplotlib, Pandas, Polars, SkLearn, Computer Vision (CV), Natural Language Processing (NLP), LLMs, Retrieval-Augmented Generation (RAG), Agentic Workflows (AutoGen, Pydantic AI) Fine-tuning, Model Evaluation

Development Django, Flask, FastAPI, Streamlit, Node.js, React, API Integration, Postman, HTML, CSS

DevOps Docker, Portainer, CI/CD, Git, Bash, Linux

Certificates

- Google Data Analytics Professional Certificate (Coursera) [🔗](#)
- Data Scientist with Python Track (DataCamp) [🔗](#)
- Python for Data Science and Machine Learning Bootcamp (Udemy) [🔗](#)